Large Language Models

From Transformers to LLMs

Stephanie Brandl

Agenda

- Midterm evaluation
- Recap: Transformers
- Decoding - How to get from word probabilities to the next token
- Training LLMs
    1. Pre-training
    2. Instruction-tuning
    3. Preference alignment

Resources

This lecture is based on

"Speech and Language Processing", Aug. 2025. Jurafsky & Martin

Chapters: Large Language Models & Post-training: Instruction tuning[...]

"Illustrating RLHF"

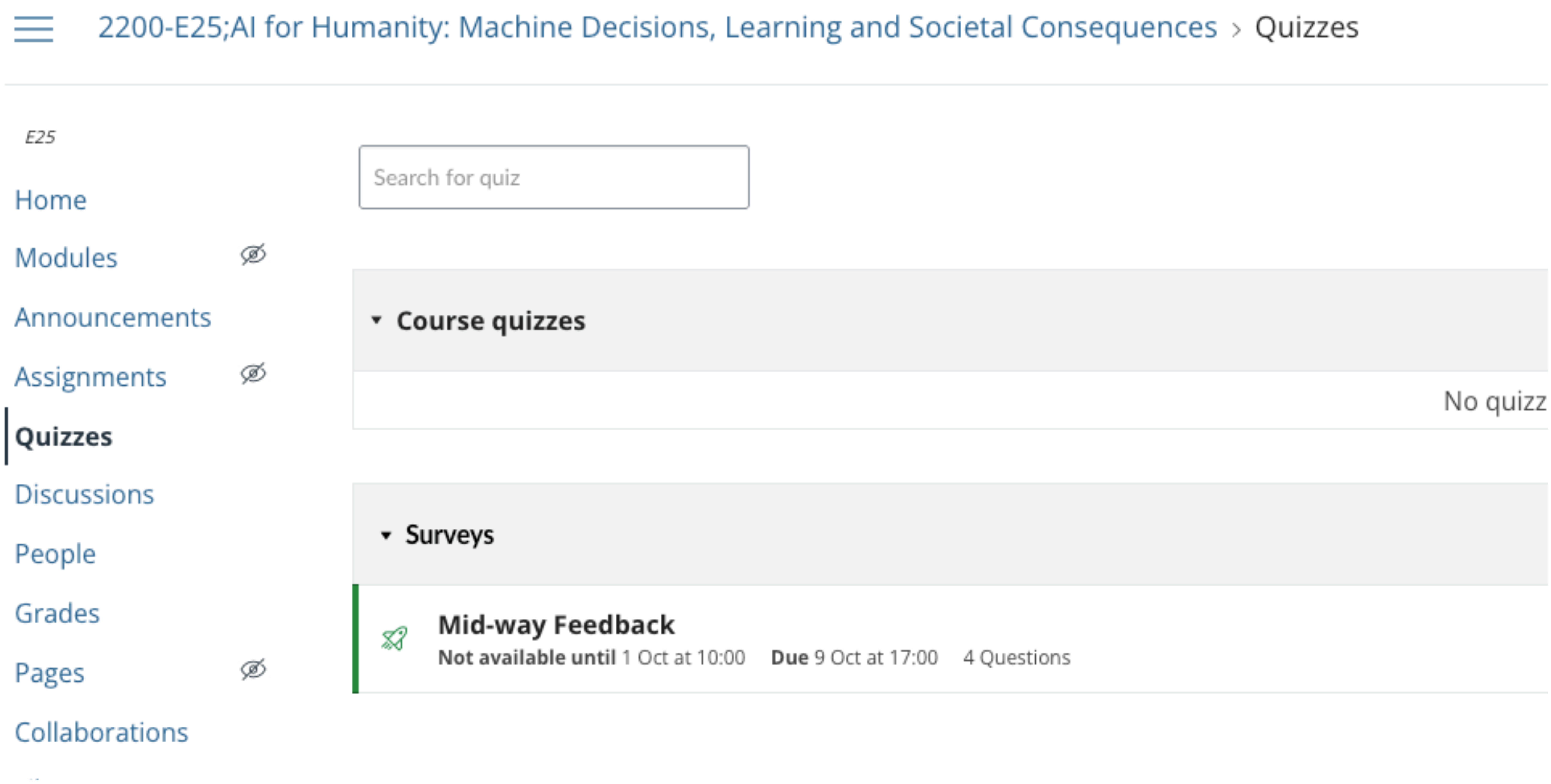
Additional resources:

"Training language models to follow instructions with human feedback" [Ouyang et al, 2022] (original RLHF paper)

"Fine-Grained Human Feedback Gives Better Rewards for Language Model Training" [Wu et al, 2023]

Midterm Evaluation

Please fill in by the end of the week



Recap

The Transformer Architecture

Encoder-Decoder

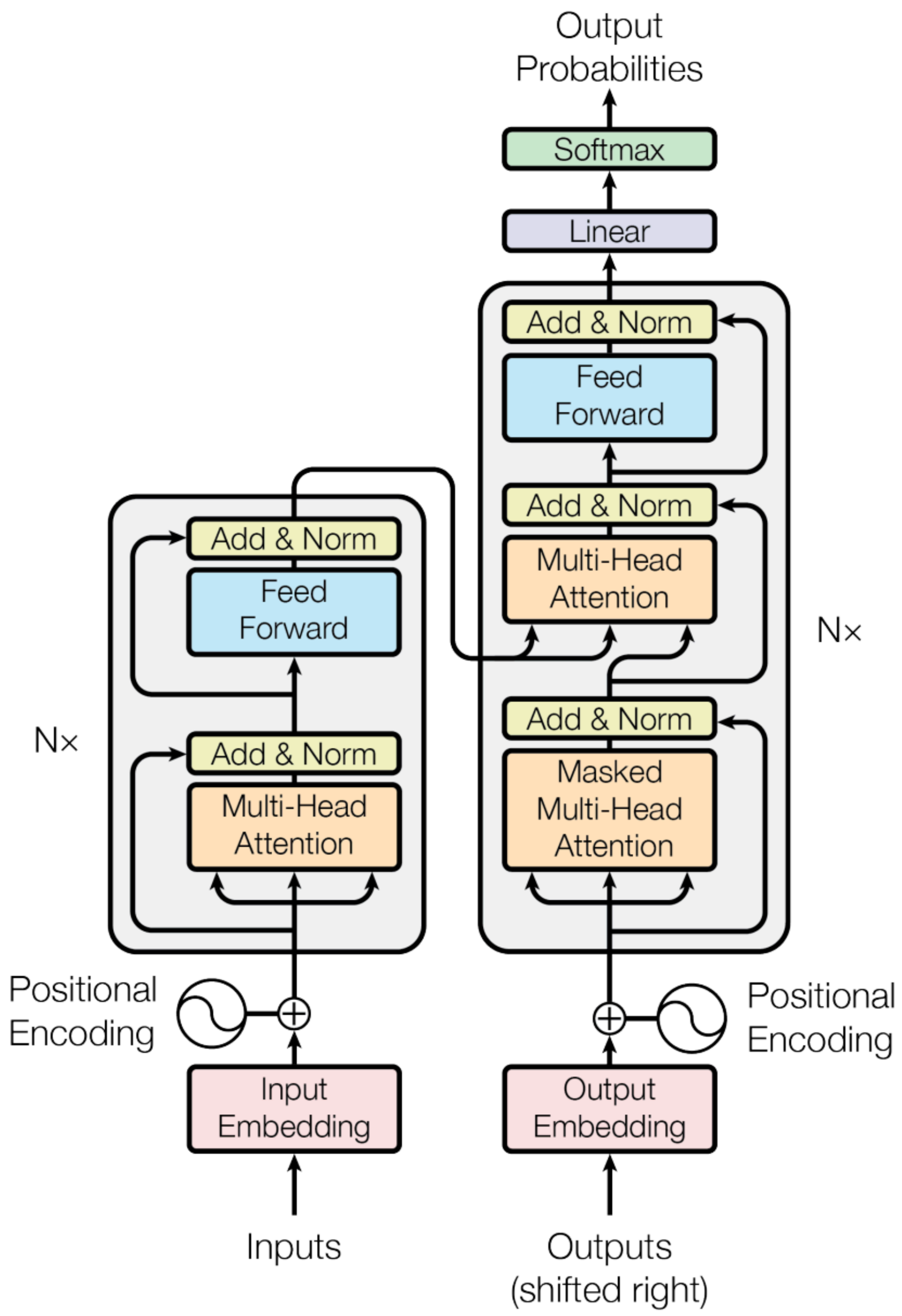


Figure 1: The Transformer - model architecture.

- Introduced in 2017 by Vaswani et al.
- Rose to prominence through BERT and GPT
- Initiated the era of pre-trained language models
- Transformers are still used in LLMs like GPT4 or Llama

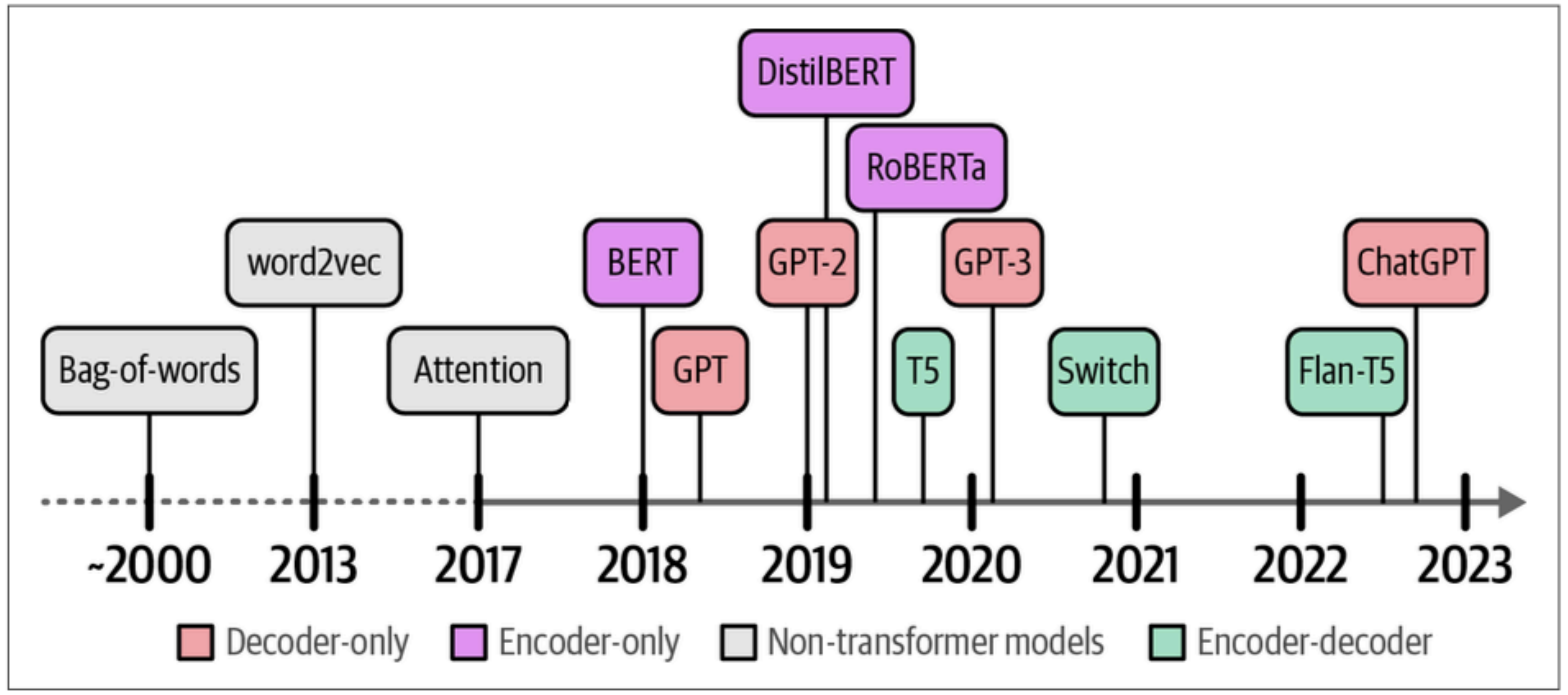


Figure 1-1. A peek into the history of Language AI.

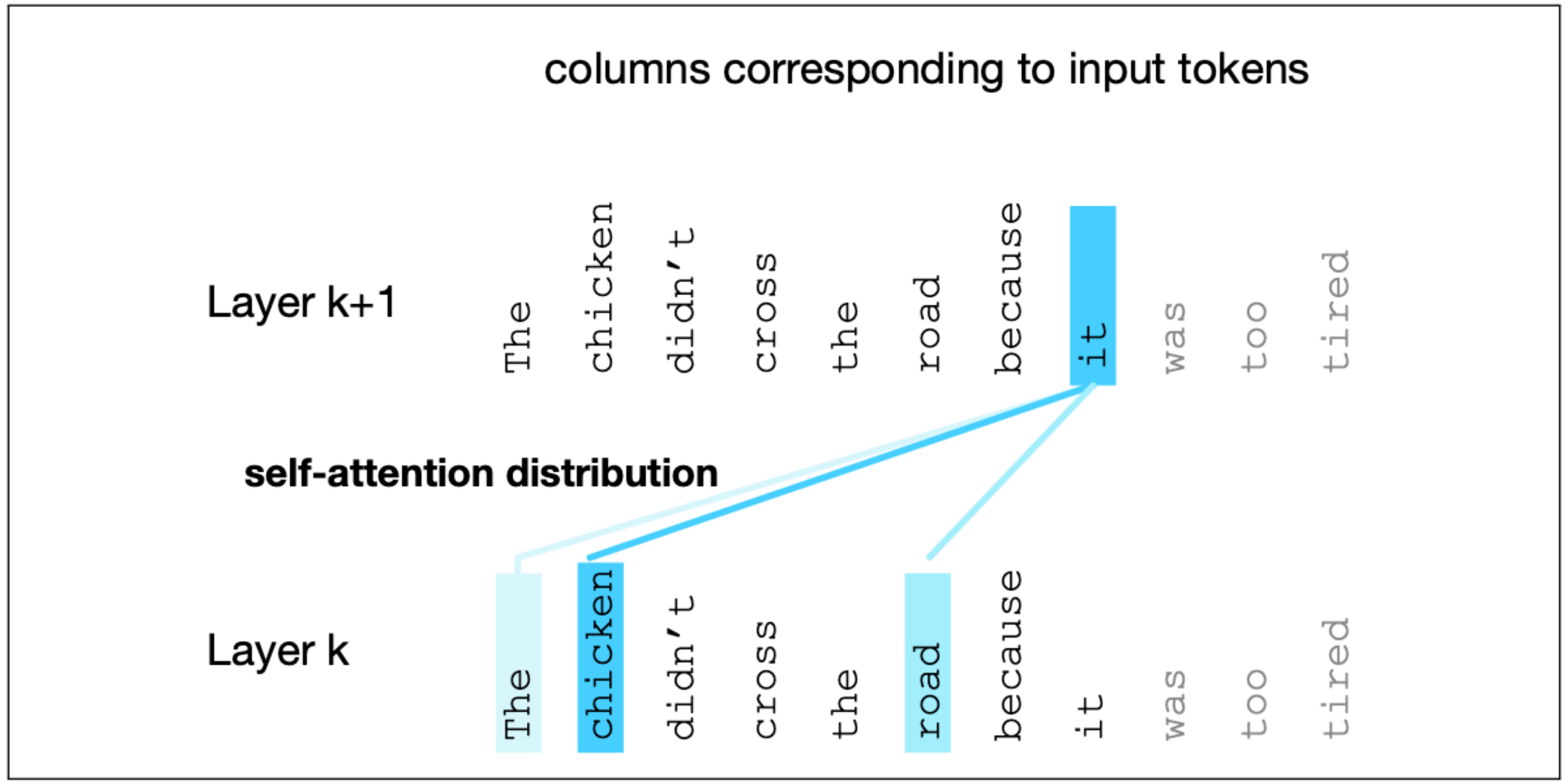
Attention Is All You Need (Vaswani et al., NeurIPS 2017)
Improving language understanding by generative pre-training (Radford et al., 2018)
BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding (Devlin et al., NAACL 2019)

Contextual Word Embeddings

- Contextual words, i.e., the words around a target word, help us to understand or compute the meaning
  - The *chicken* didn't cross the road because *it* was too *tired*
  - The chicken didn't cross the *road* because *it* was too *wide*
- Relevant contextual information can be "far" away from the target word, remember that we used a window size of 5 words for word2vec
- Transformers are able to build contextual word embeddings by integrating the meaning of relevant contextual words
- layer by layer, richer and richer contextual representations of the input words are learned
- at each layer, the representation of a word is updated by combining information from previous layers with information about neighbouring tokens

Self-attention

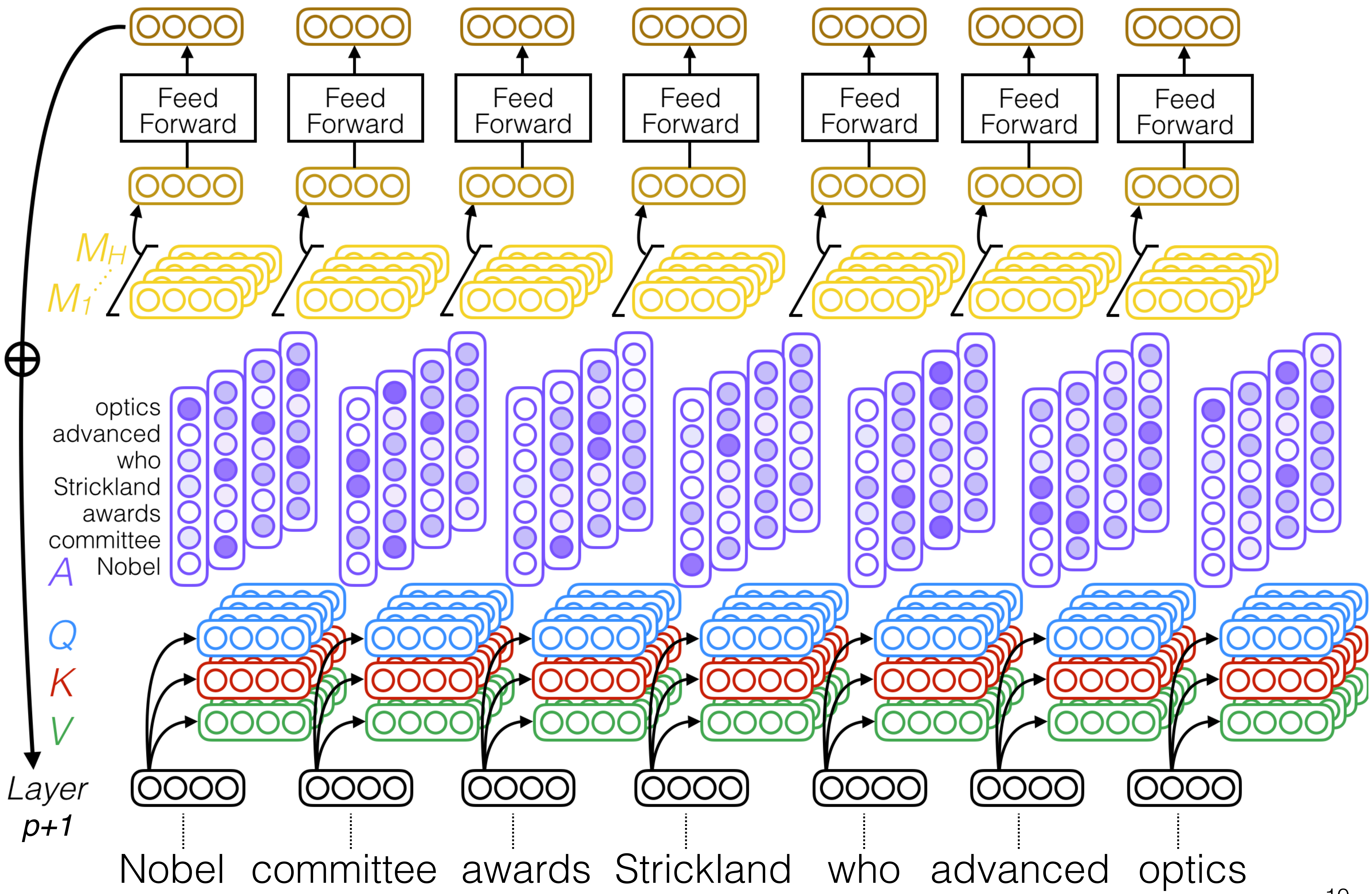
- self-attention is the mechanism behind contextual word embeddings
- it weighs and combines the representation of relevant tokens in the context from layer $k-1$ to build the representation for tokens in layer k
- it allows tokens to "communicate" with each other



Example for attention weights

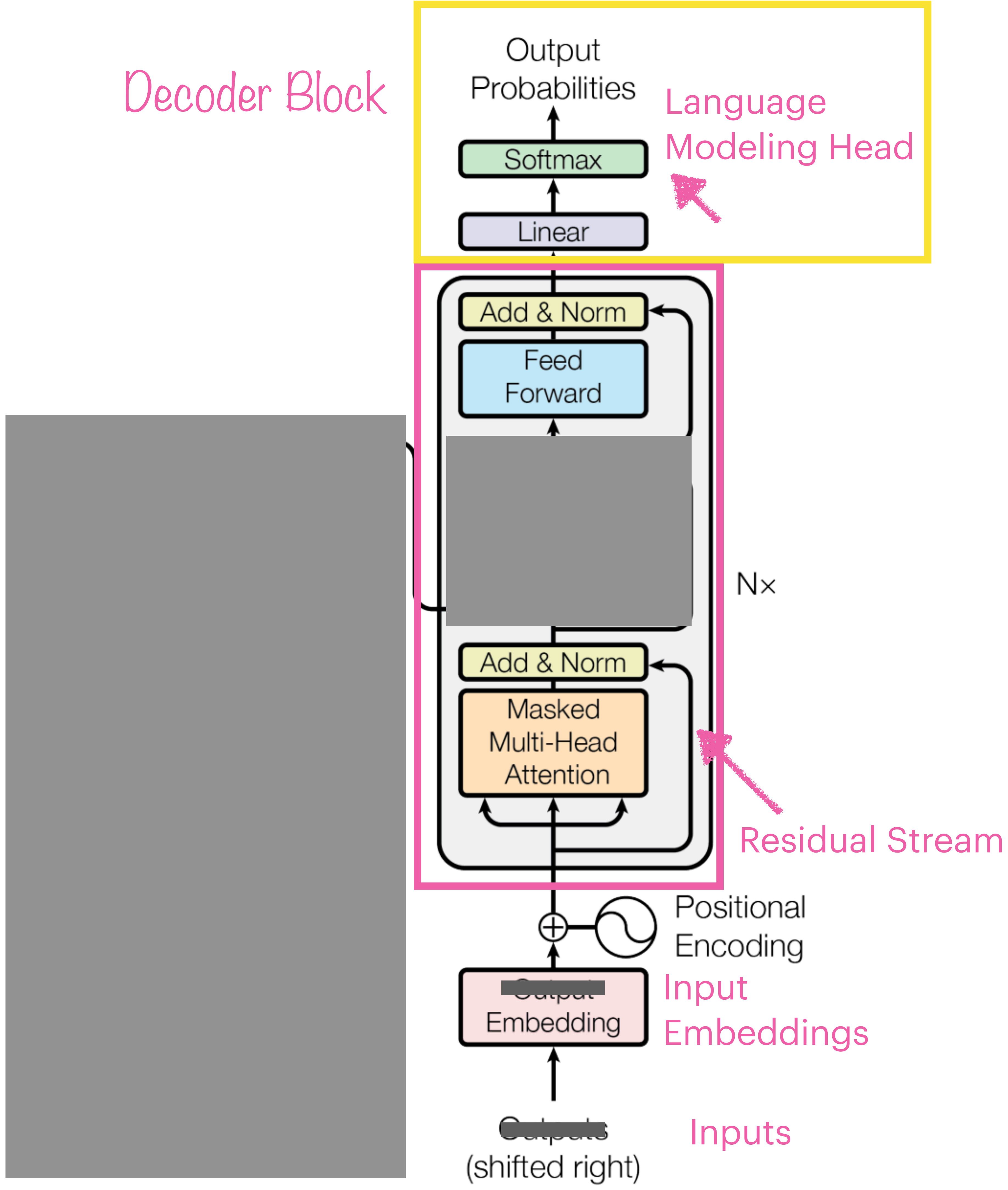
[Vaswani et al. 2017]

Multi-head self-attention



Decoding

Predicting the next token



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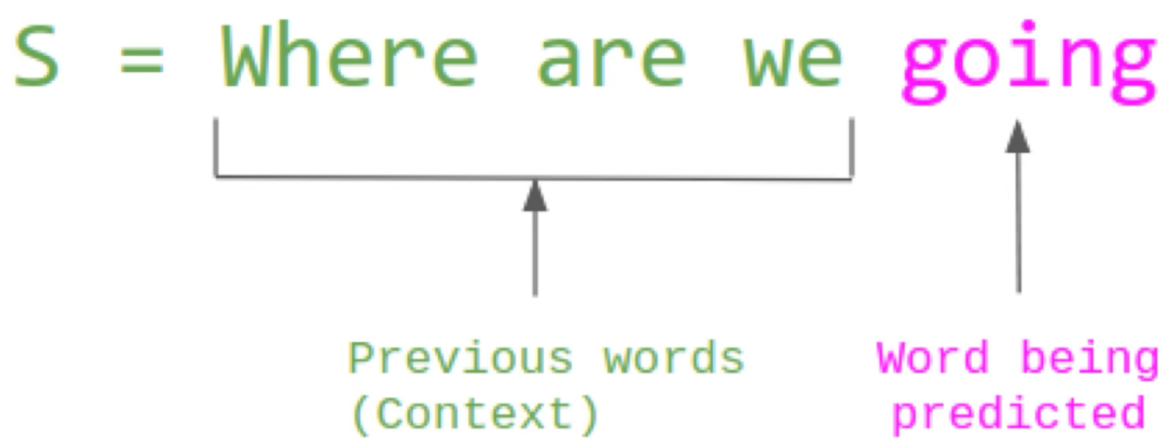


P(S) = P(Where) x P(are | Where) x P(we | Where are) x P(going | Where are we)

Reminder of the main LM task

How do we get from the output probabilities to the next token?

Decoding

Decoding: the task of choosing a next token based on the model's output probabilities

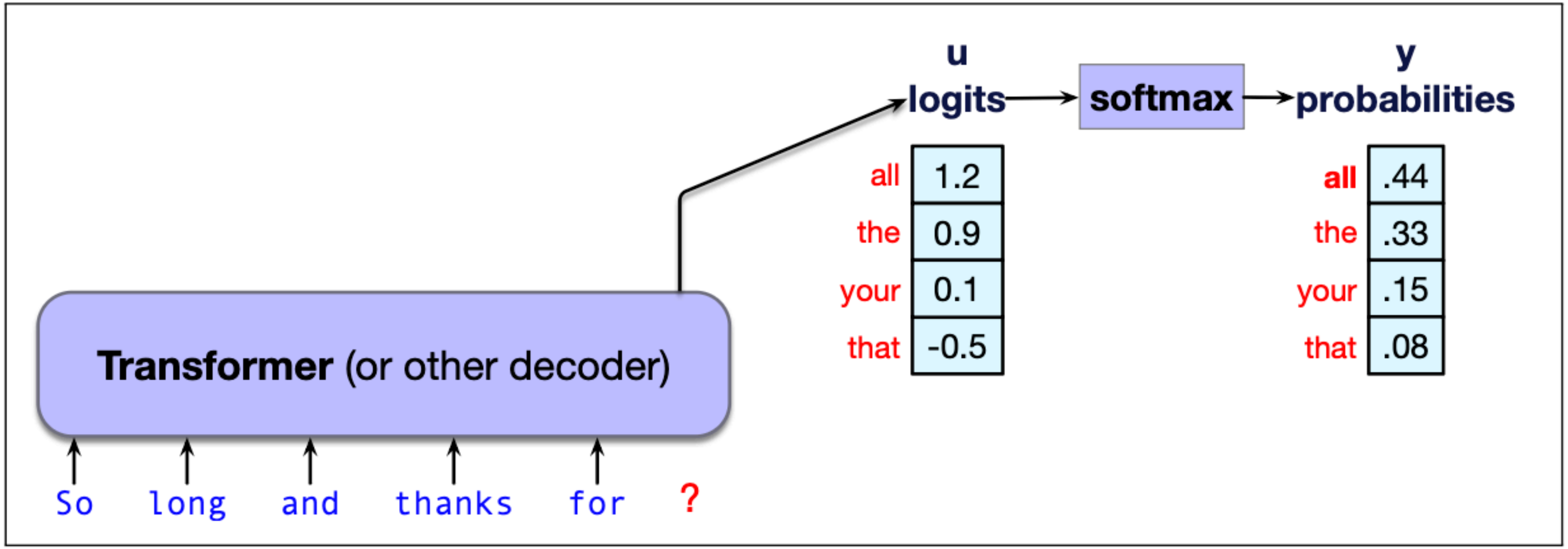


Figure 7.7 Taking the logit vector **u** and using the softmax to create a probability vector **y**.

from Large Language Models [Jurafsky & Martin, Aug. 25]

Naive approach: Greedy decoding

- Greedy decoding: the simplest way to generate tokens is to always choose the most likely token given the output probabilities

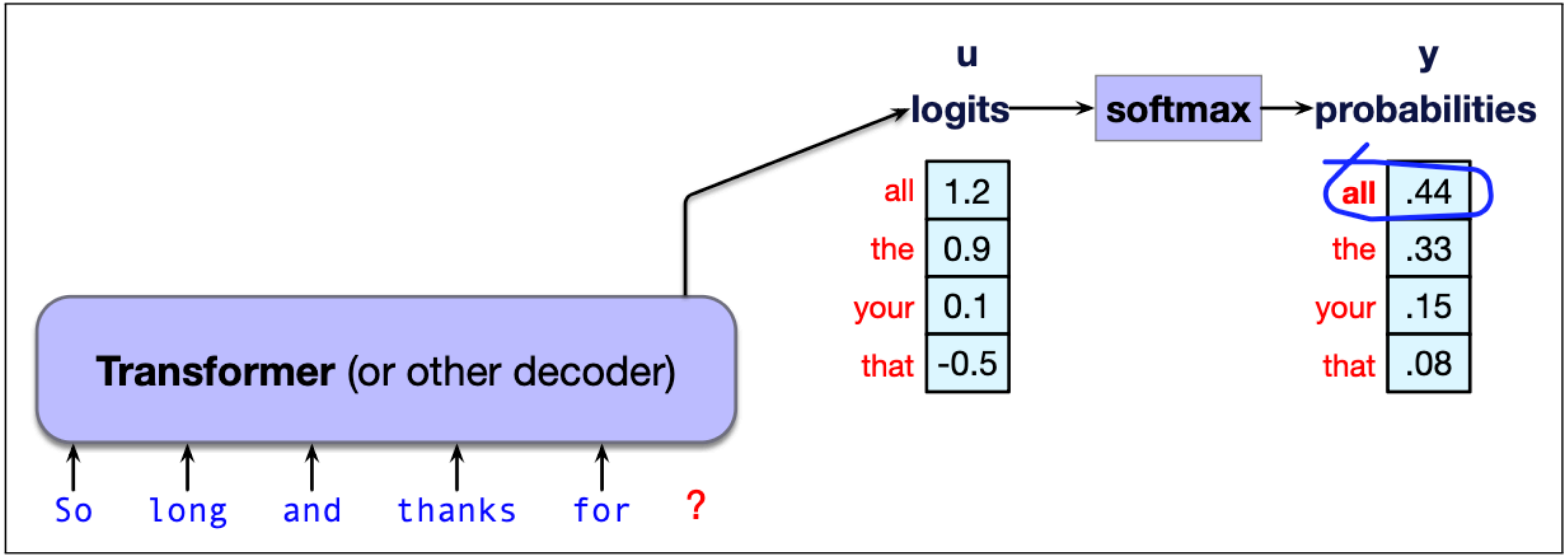


Figure 7.8 Greedy decoding: choose the highest probability word.

from Large Language Models [Jurafsky & Martin, Aug. 25]

Naive approach: Greedy decoding

- Greedy decoding: the simplest way to generate tokens is to always choose the most likely token given the output probabilities

⚠️ Why this might not be a good idea ⚠️

- tokens are by definition extremely predictable
- the generated text is generic and repetitive
- even deterministic: for identical context the same model will always generate the same text

→ sampling methods

Random Sampling

- Sampling from the distribution of output probabilities
  - Words with higher probabilities are more likely to be chosen
  - Words with lower probabilities still have a (smaller) chance

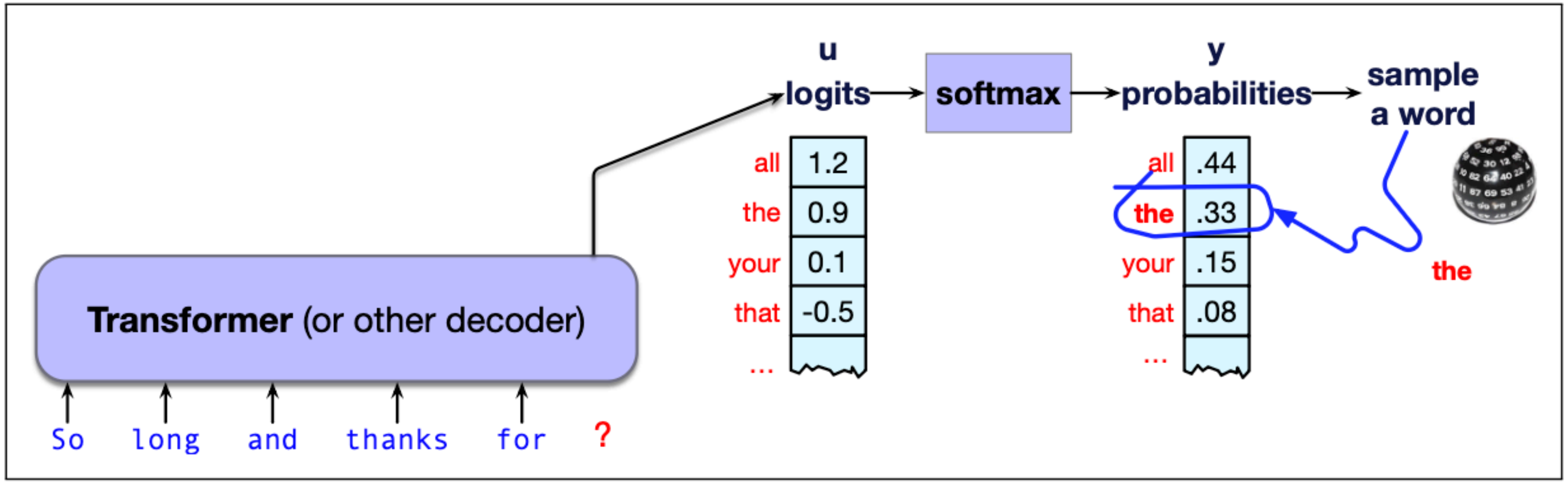


Figure 7.9 Random multinomial sampling: we randomly chose a word according to its probability.

from Large Language Models [Jurafsky & Martin, Aug. 25]

Random Sampling

- Sampling from the distribution of output probabilities
  - Words with higher probabilities are more likely to be chosen
  - Words with lower probabilities still have a (smaller) chance

⚠️ Why this might not be a good idea ⚠️

- mostly generates sensible, high-probable tokens
- but still there are too many odd, rare tokens in the mix

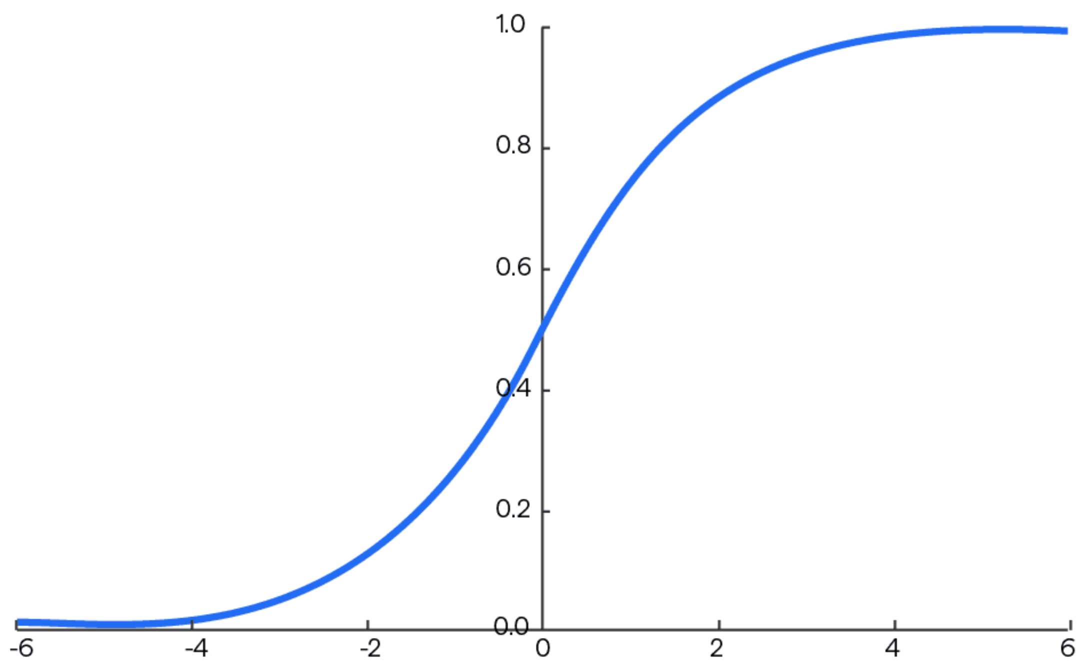
→ temperature sampling

Temperature Sampling

- Reshaping the probability distribution to increase the probabilities of high probability tokens and decrease the probabilities of low probability tokens

- We scale the output probability vector $\mathbf{o}$ with the temperature parameter $\tau \in (0,1]$

Softmax Function



$$\mathbf{y} = \text{softmax}(\mathbf{o}/\tau)$$

💥 Why does this work? 💥

- if $\tau = 1$, the probabilities stay the same

- if $\tau \leq 1$, all probabilities inside the softmax increase

- softmax tends to push high values towards 1 and low values towards 0

Temperature Sampling

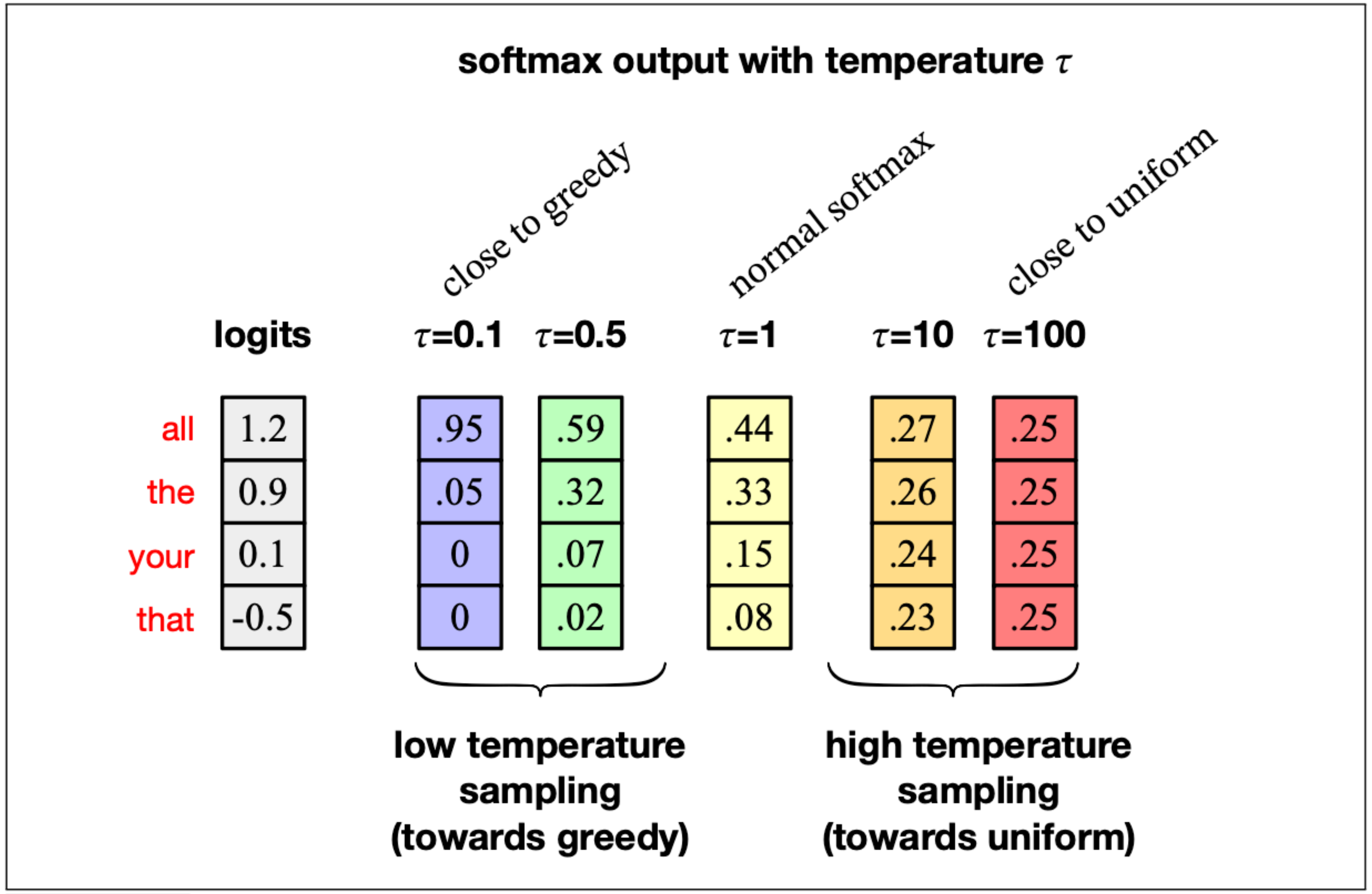


Figure 7.11 Seeing how different values of τ change the resulting probabilities from the initial logits in temperature sampling. In this simplified example, there are only 4 tokens in the vocabulary.

from Large Language Models [Jurafsky & Martin, Aug. 25]

Let’s take one step back

Where are we right now?

- Language Models ✅
- Transformer Architecture ✅
- Predicting next words ✅
- Large language models? 🤨
- Training? 🤯
- Data? 🙀

Large Language Models

How do we got from Language Models to Large Language Models?

- There is no clear definition, some ideas
  - an LLM is just a much larger version of an LM
  - an LLM has more capabilities, e.g., is able to follow instructions
- How does an LLM learn to follow instructions?
  - instruction-tuning, hang in there, we will get there!
- What is considered an LLM changes quickly

> Note: Given the size of the language models used in this study (all models <= 13B), I encourage the authors to follow the literature and call them small language models (SLMs) instead of LLMs.

from a review I recently received

Training an LLM

The three stages of training

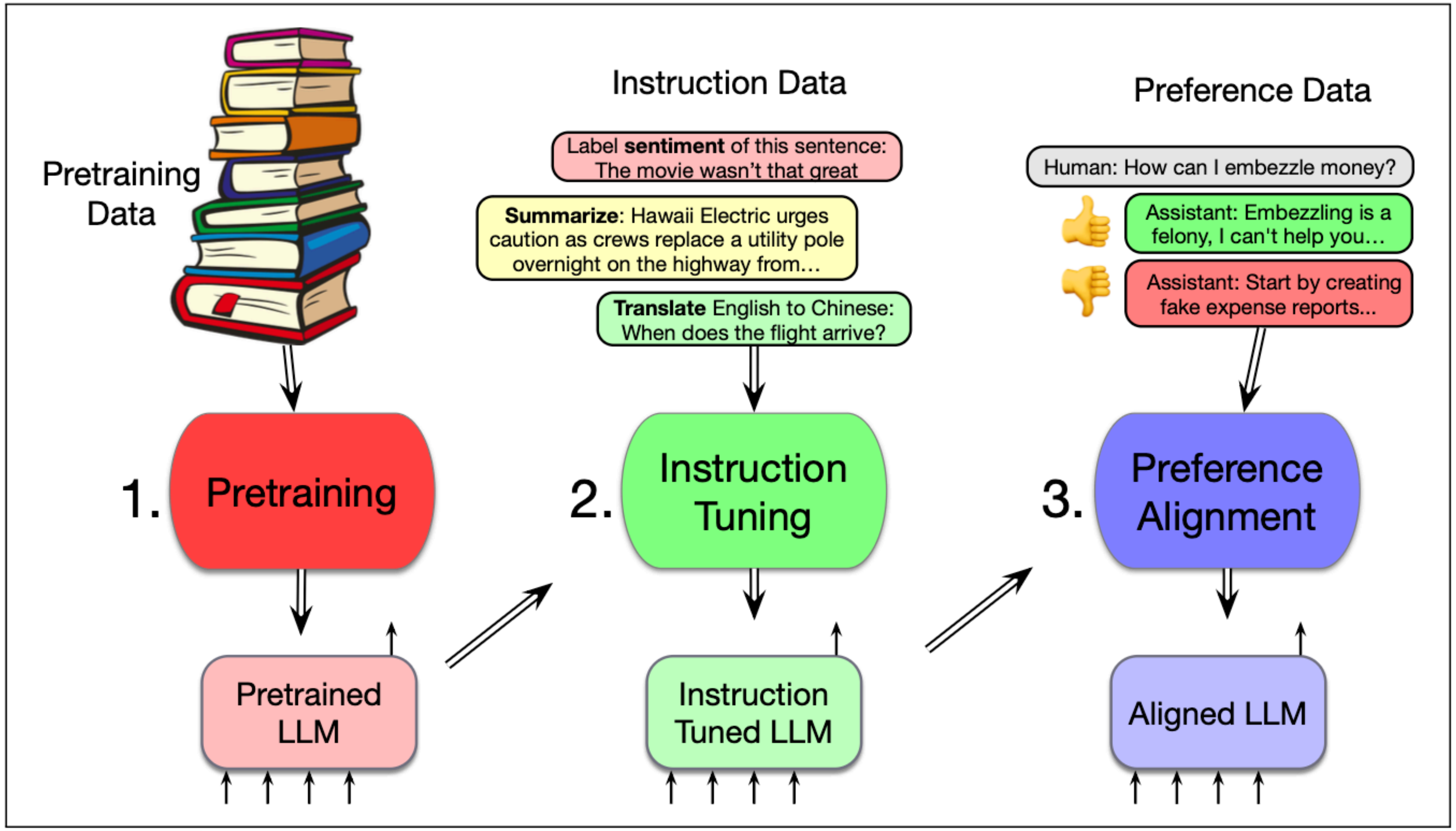


Figure 7.12 Three stages of training large language models: pretraining, instruction tuning, and preference alignment.

from Large Language Models [Jurafsky & Martin, Aug. 25]

1. Pre-training

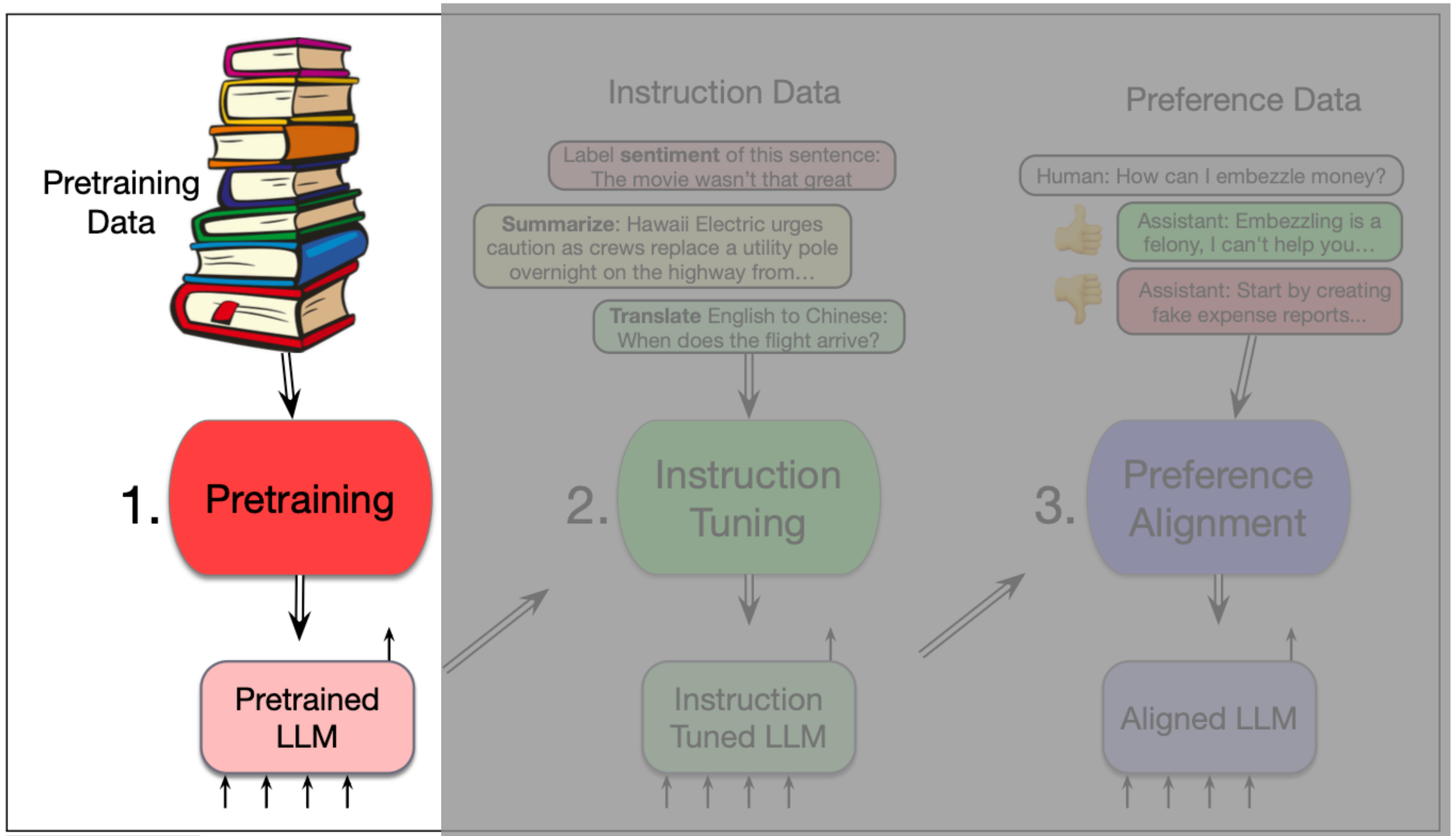


Figure 7.12 Three stages of training large language models: pretraining, instruction tuning, and preference alignment.

- model is trained from scratch on next token prediction
- optimization based on cross-entropy loss
- training data is usually an enormous web corpus

Result: a model that can generate text and is good at predicting words

from Large Language Models [Jurafsky & Martin, Aug. 25]

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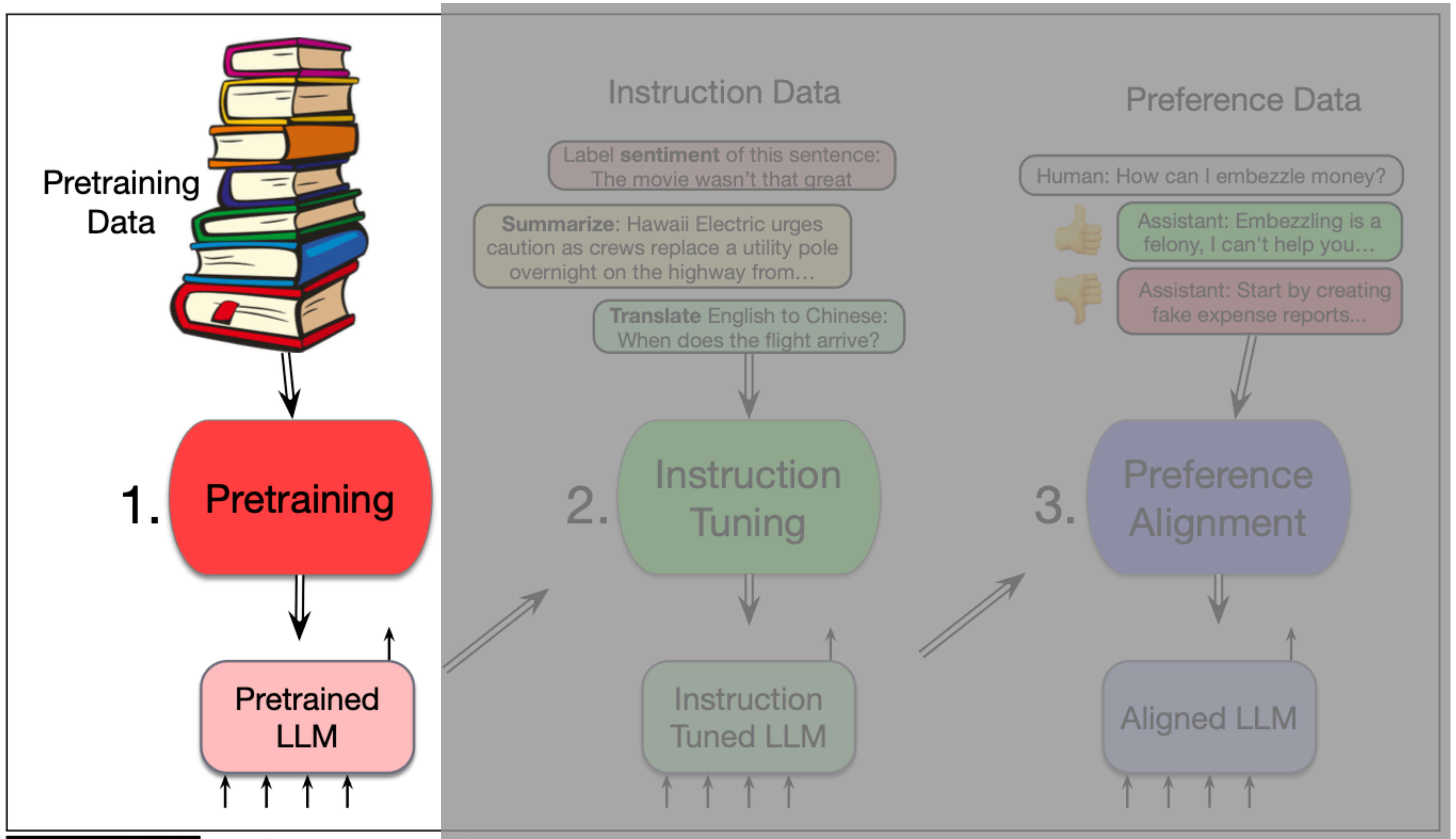

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Why next token prediction?

- What can language models learn from next token prediction?

With roses, dahlias, and peonies, I was surrounded by ______ flowers
The room wasn't just big it was ______ enormous
The square root of 4 is ______ 2
The author of "A Room of One's Own" is ______ Virginia Woolf
The professor said that ______ he

Why next token prediction?

- What can language models learn from next token prediction?

With roses, dahlias, and peonies, I was surrounded by ______ flowers	(ontological) facts
The room wasn't just big it was ______ enormous	
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Why next token prediction?

- What can language models learn from next token prediction?

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The room wasn't just big it was ______	enormous	enormous >> big
The square root of 4 is ______	2	
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The professor said that ______	he	

Why next token prediction?

- What can language models learn from next token prediction?

With roses, dahlias, and peonies, I was surrounded by ______ flowers	(ontological) facts
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The square root of 4 is ______ 2	some math
The author of "A Room of One's Own" is ______ Virginia Woolf	(world) facts
The professor said that ______ he	possible bias

Self-supervised pretraining

Self-supervised training: similar to word2vec, we take a corpus of text and ask the model to predict the next word at each time step t.

- since we know the correct answer (the next word in the corpus), we can optimize and update the model to get better at predicting next tokens
- for this we need a loss function: cross-entropy loss

$$L_{CE}(\hat{\mathbf{y}}_t, \mathbf{y}_t) = -\log \hat{\mathbf{y}}_t[w_{t+1}]$$

where $\hat{\mathbf{y}}$ represents the estimated word probability vector

for more on that: Ch. 7.5 and Ch. 4.6 of Jurafsky & Martin (Aug. 2025) and Lecture 2

- teacher forcing: we always give the model the correct history sequence, ignoring potential mistakes that happen before
- remember backpropagation from lecture 2

Pretraining data

- mainly text scraped from the web
- likely contains many natural examples that are helpful for NLP taks
    - question-answer pairs
    - translations
    - documents with their summaries
- Common Crawl: a series of snapshots from the web,
    - for instance C4: 156 billion English tokens, mostly patent text documents, Wikipedia and news sites
- The Pile: 825 GB English text corpus
    - publicly released code, text from the web, books and Wikipedia

Pretraining data

Filtering:

- quality:
  - avoid websites with personal identifiable information or adult content
  - deduplication
- safety
  - toxicity detection

Ethical and legal questions:

- Copyright
- Data consent
- Privacy
- Skew/Bias

Pretraining data

Meta wins AI copyright lawsuit as US judge rules against authors

Writers accused Facebook owner of breach over its use of books without permission to train its AI system

A Meta spokesperson called fair use a 'vital legal framework' for building 'transformative' AI technology. Photograph: Dado Ruvić/Reuters

Anthropic wins key US ruling on AI training in authors' copyright lawsuit

By Blake Brittain

June 24, 2025 7:50 PM GMT+2 · Updated June 24, 2025

Aa

Anthropic logo is seen in this illustration taken March 31, 2023. REUTERS/Dado Ruvic/Illustration/File Photo Purchase Licensing Rights

So far, so good?

Prompt: Explain the moon landing to a six year old in a few sentences.
Output: Explain the theory of gravity to a 6 year old.

Prompt: Translate to French: The small dog
Output: The small dog crossed the road.

- the capabilities of a pre-trained LLM are limited
- not sufficiently helpful
  - instruction-tuning
- potentially harmful
  - alignment

2. Instruction (Fine)Tuning

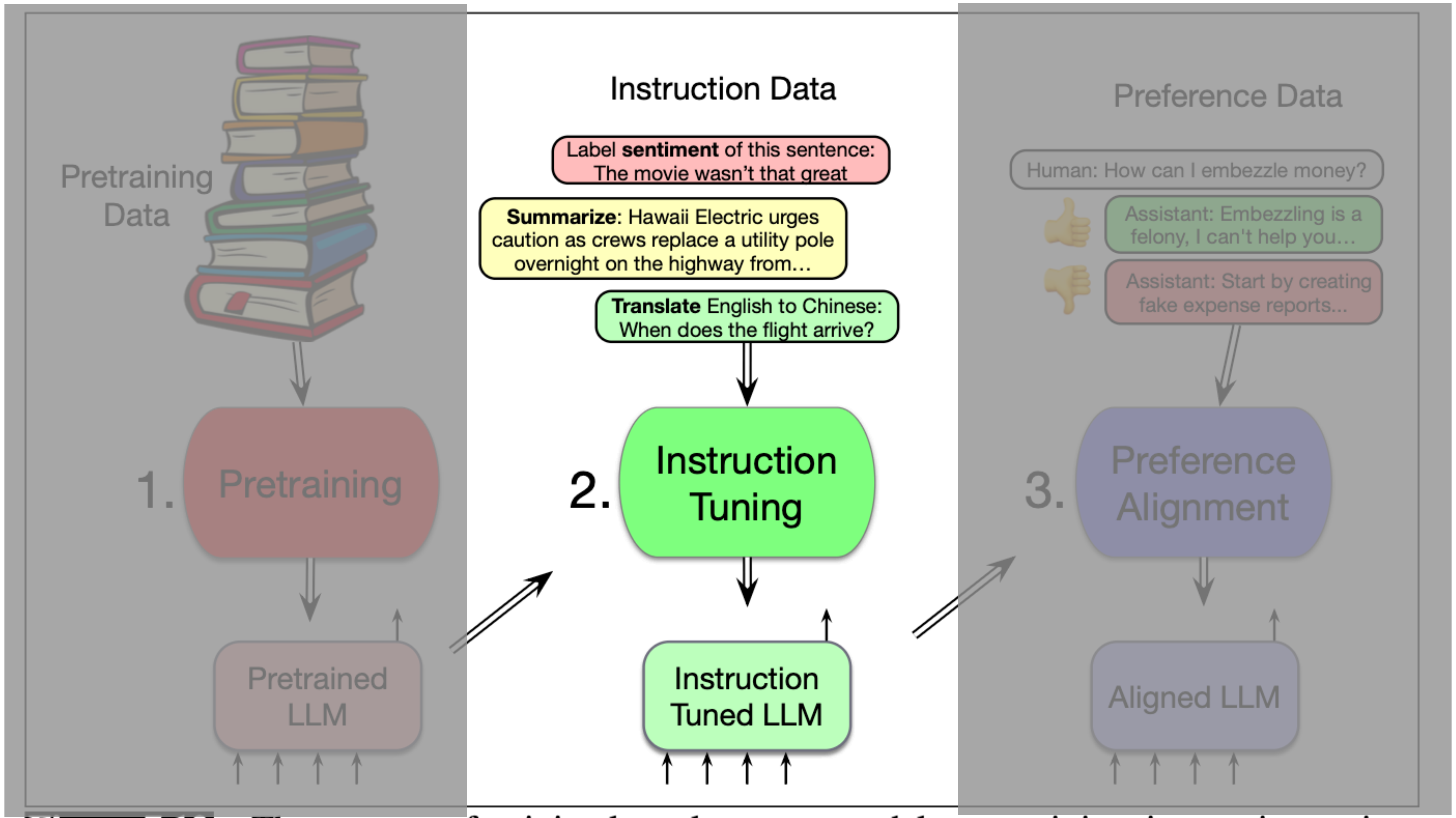


Figure 7.12 Three stages of training large language models: pretraining, instruction tuning, and preference alignment.

from Large Language Models [Jurafsky & Martin, Aug. 25]

2. Instruction (Fine)Tuning

- 🫡 to make the model better at following instructions
- 🔁 take a base **pretrained** model and train it to follow instructions
- ❓ training corpus of instructions and responses
- 🤹 **meta- learning**: the model learns to follow instructions in general
- 📝 same language model objective as before: <u>next token prediction</u>
- 🕵️ **supervised learning**: every instruction has a correct answer
- 💰 cost of instruction tuning is a small fraction of the original training cost since datasets are much smaller

What do we mean by instructions?

- natural language description of a task combined with labeled task demonstrations
  - Answer the following question
  - Translate the following text to [TARGET LANGUAGE]
  - Summarize this report
- can include
  - length restrictions
  - personas to assume
  - requirements for a specific file format

Aya Dataset

por	Qual é a capital do estado da Bahia?	A capital do estado da Bahia, no Brasil, é Salvador. Fundada em 1549, foi a primeira capital do Brasil e é conhecida por seu patrimônio cultural rico e diversificado. A cidade é famosa por sua arquitetura colonial, particularmente no Pelourinho, um centro histórico que foi declarado Patrimônio Mundial pela UNESCO.
pes	در برج کبوترخانه میبد چند لانه کبوتر وجود دارد؟	کبوترخانه میبد برجی استوانه‌ای است که داخل آن چهارهزار لانه کبوتر وجود دارد و نوار سفید رنگی برای جذب کبوتران دورتادور برج کشیده شده است. این برج در برابر خطر حمله دیگر حیوانات به کبوترها بسیار ایمن بوده است.
msa	Apakah nasi lemak?	Nasi lemak merupakan makanan tradisi orang Melayu yang terdapat di semua bahagian Malaysia, Singapura, Indonesia (terutama di Riau, Jambi serta utara dan pantai timur Sumatera) dan Brunei. Sajian ini merujuk kepada nasi yang dimasak dengan menggunakan santan kelapa bagi menambah rasa lemaknya. Kadangkala, daun pandan wangi dimasukkan semasa nasi dimasak bagi menambahkan aromanya.
tam	செயற்கை நுண்ணறிவு என்றால் என்ன?	பொதுவாக மனிதர்களால் செய்யப்படும் பணிகளைச் செய்ய ஒரு கணினி அல்லது ஒரு கணினியால் கட்டுப்படுத்தப்படும் ஒரு ரோபோவின் திறன் செயற்கை நுண்ணறிவு எனப்படும்.

Figure 10.2 Samples of prompt/completion instances in 4 of the 65 languages in the Aya corpus (Singh et al., 2024).

- participatory research initiative
- 503 million instructions
- 114 languages
- 12 tasks like question answering, summarization, translation, paraphrasing, sentiment analysis, natural language inference
- link to dataset

from Post-training: Instruction Tuning, Alignment and Test-Time Compute [Jurafsky & Martin, Aug. 25]

Super Natural Instructions Dataset

take data points ➡️ and plug them into templates ⬇️ to transform the original NLP format into an instruction format

Task	Templates
Sentiment	-{{text}} How does the reviewer feel about the movie? -The following movie review expresses what sentiment? {{text}} -{{text}} Did the reviewer enjoy the movie?
Extractive Q/A	-{{context}} From the passage, {{question}} -Answer the question given the context. Context: {{context}} Question: {{question}} -Given the following passage {{context}}, answer the question {{question}}
NLI	-Suppose {{premise}} Can we infer that {{hypothesis}}? Yes, no, or maybe? -{{premise}} Based on the previous passage, is it true that {{hypothesis}}? Yes, no, or maybe? -Given {{premise}} Should we assume that {{hypothesis}} is true? Yes,no, or maybe?

Figure 10.4 Instruction templates for sentiment, Q/A and NLI tasks.

Few-Shot Learning for QA

Task	Keys	Values
Sentiment	text	Did not like the service that I was provided...
	label	0
	text	It sounds like a great plot, the actors are first grade, and...
	label	1
NLI	premise	No weapons of mass destruction found in Iraq yet.
	hypothesis	Weapons of mass destruction found in Iraq.
	label	2
	premise	Jimmy Smith... played college football at University of Colorado.
	hypothesis	The University of Colorado has a college football team.
	label	0
Extractive Q/A	context	Beyoncé Giselle Knowles-Carter is an American singer...
	question	When did Beyoncé start becoming popular?
	answers	{ text: ['in the late 1990s'], answer_start: 269 }

Figure 10.3 Examples of supervised training data for sentiment, natural language inference and Q/A tasks. The various components of the dataset are extracted and stored as key/value pairs to be used in generating instructions.

from Post-training: Instruction Tuning, Alignment and Test-Time Compute [Jurafsky & Martin, Aug. 25]

3. Preference-based Learning

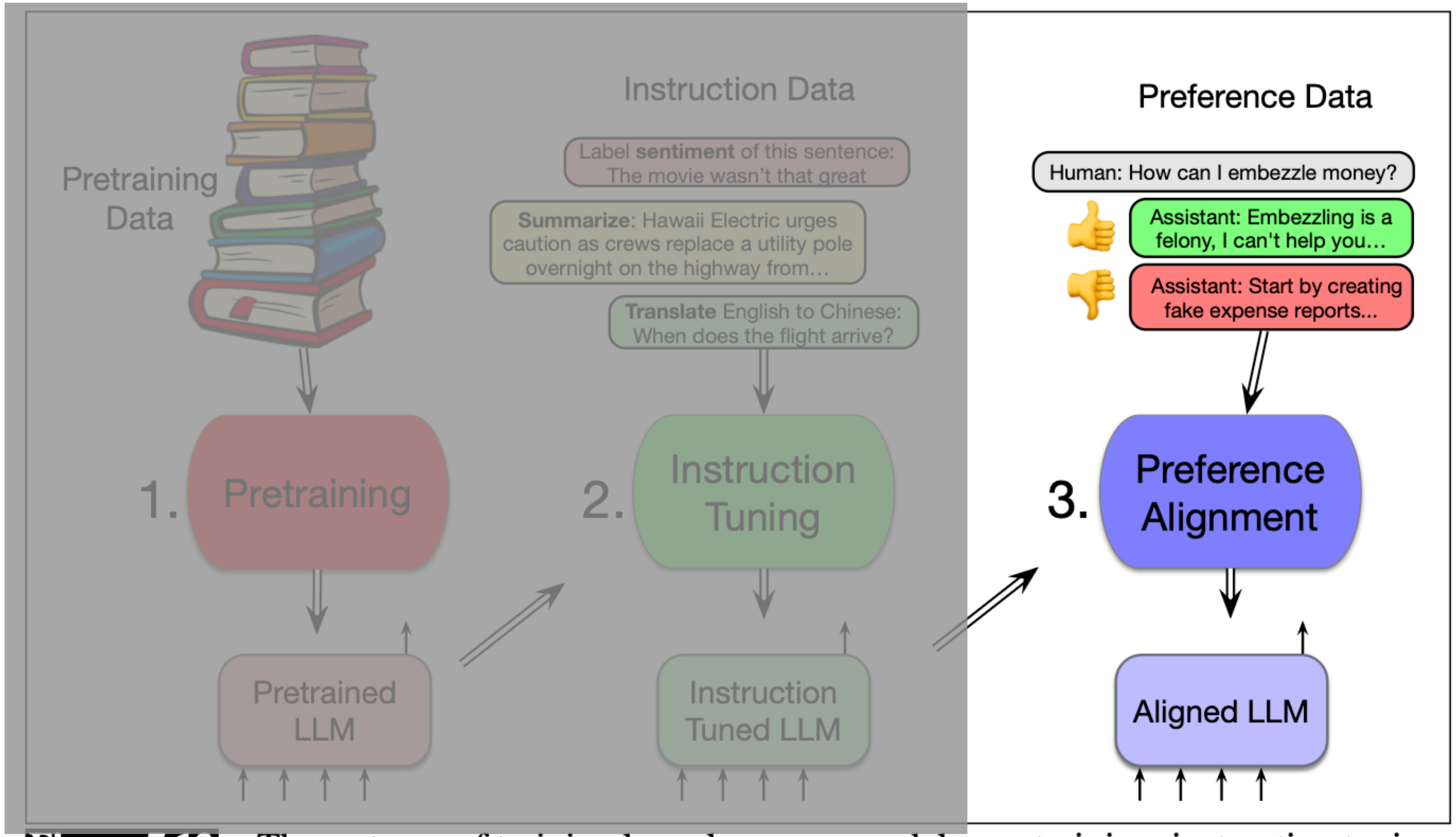


Figure 7.12 Three stages of training large language models: pretraining, instruction tuning, and preference alignment.

from Large Language Models [Jurafsky & Martin, Aug. 25]

3. Preference-based Learning

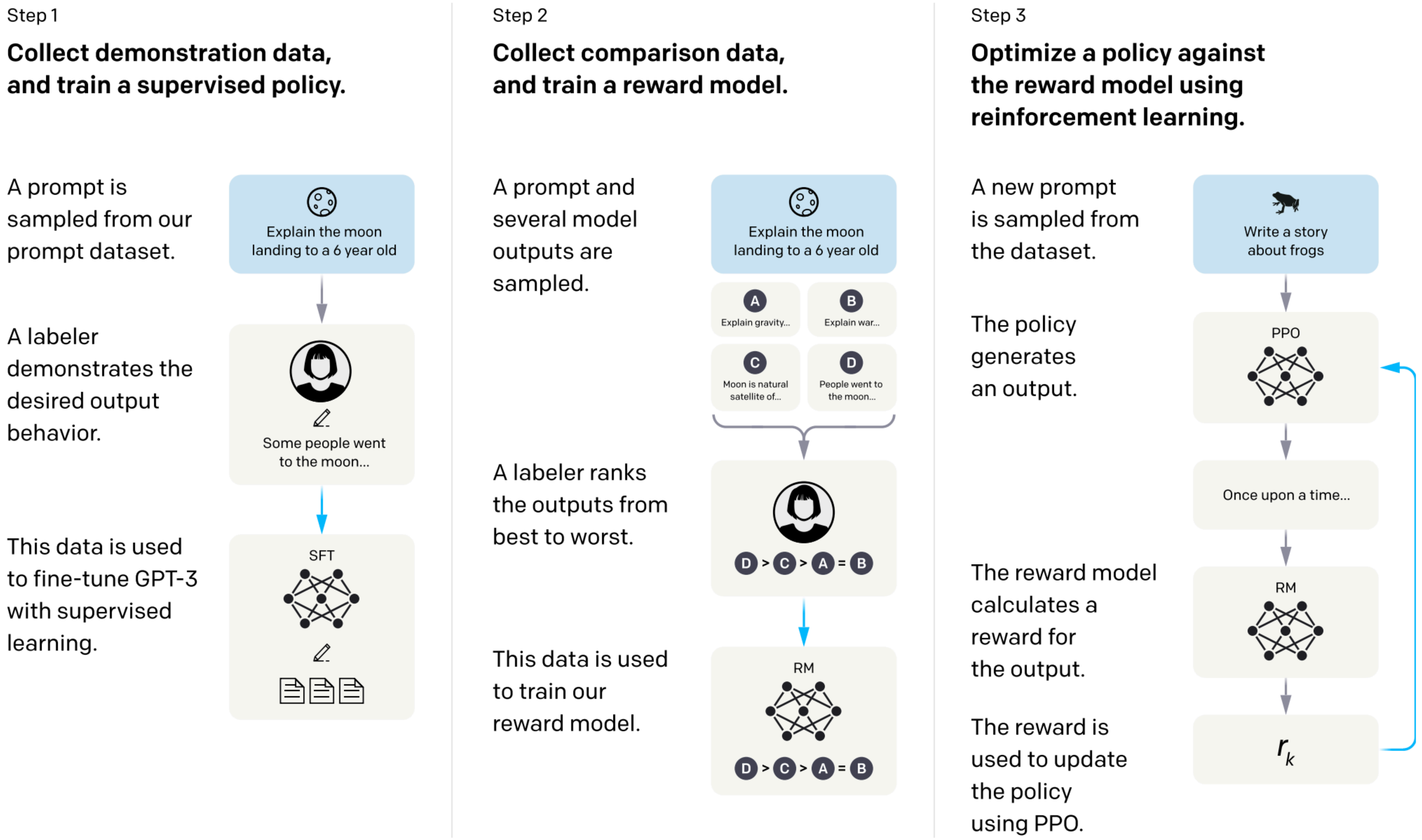
- with instruction tuning the models learns to follow instructions
- does not prevent the model from problematic behaviour like:
  - hallucinations, unsafe, harmful, toxic outputs
- preference-based learning or “alignment” aims to improve:

Figure 1: The three pillars of alignment in the examined work: *helpfulness*, *harmlessness*, *truthfulness*, and the core underlying aspects they aim to promote or prevent.

from [Chalkidis, 2025]

3. Preference-based Learning

Reinforcement Learning from Human Feedback (RLHF)

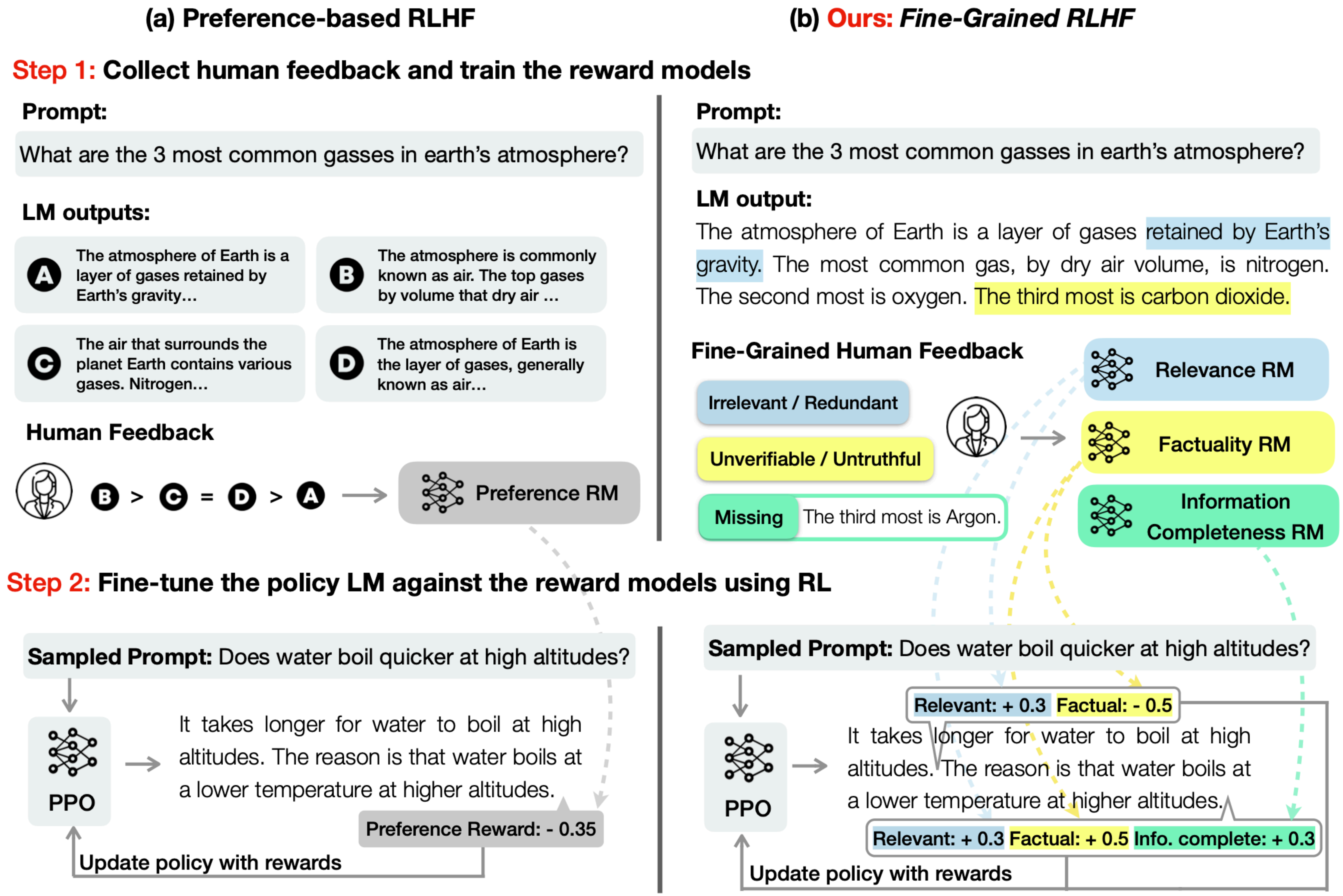


from [Ouyang et al., 2022]

Reinforcement Learning from Human Feedback (RLHF)

- Step 1: Instruction-tune LLM
- Step 2: Train a reward model
  - a model that takes in a sequence of text (prompt, response) and outputs a reward score
  - based on prior ranking of LLM outputs
  - reward model can be another LLM itself
- Step 3: Policy optimization
  - the reward model is used to update the instruction-tuned LLM
  - the proximal policy optimization (PPO) framework combines the reward model and the LLM with a constraint on policy shift

Fine-grained RLHF



from [Wu et al., 2023]

All problems solved?

Crowd workers

'It's destroyed me completely': Kenyan moderators decry toll of training of AI models

Employees describe the psychological trauma of reading and viewing graphic content, low pay and abrupt dismissals

Link to Guardian

BUSINESS • TECHNOLOGY

Exclusive: OpenAI Used Kenyan Workers on Less Than $2 Per Hour to Make ChatGPT Less Toxic

Link to Time

- someone has to do the ranking for the RLHF framework
- annotators come predominantly from the global South
- are paid very poorly
- work under horrendous conditions to rate toxic content generated by LLMs

Who decides what a “good” policy is?

10 July 2023

ChatGPT promotes American norms and values

TECHNOLOGY | ARTIFICIAL INTELLIGENCE

ARTIFICIAL INTELLIGENCE ChatGPT, the revolutionary new AI chatbot, reflects American norms and values – even when queried about other countries and cultures. The mismatch has been demonstrated in research from the University of Copenhagen. The AI spun web of cultural bias is a major problem according to the study’s researchers.

DIKU News

- The rankings in RLHF follow pre-defined guidelines that contain preferences or values
- Often unclear what those values are
- Who came up with them
- Subjective
- Could also steer the LLM into the “wrong” direction

Safety filters are easy to "jailbreak"

Study says ChatGPT giving teens dangerous advice on drugs, alcohol and suicide

Link

A Teen Was Suicidal. ChatGPT Was the Friend He Confided In.

More people are turning to general-purpose chatbots for emotional support. At first, Adam Raine, 16, used ChatGPT for schoolwork, but then he started discussing plans to end his life.

Link to NYT

Unclear what impact LLMs have on us

Updated: JUN 23, 2025 4:16 PM CET

ChatGPT May Be Eroding Critical Thinking Skills, According to a New MIT Study

TECH | ARTIFICIAL INTELLIGENCE

Link to TIME

Conclusion

- We need sampling methods to decode the output probability of a Transformer into the next word
- Training an LLM includes 3 stages:
  - Pre-training
  - Instruction-tuning
  - Preference alignment
- Several problems arise:
  - crowd workers are underpaid and have to deal with toxic content
  - safety filters can be bypassed with horrible consequences
  - unclear what the excessive use of LLMs does to society